One Hot Encoding

Vocabulary size = 7
vocabulary (unique words)

Text	O/P	The	food	is	good	bad	Pizza	Amazing
D1 The food is good	1	1	0	0	0	0	0	0
D2 The food is bad	0	0	1	0	0	0	0	0
D3 Pizza is Amazing	1							

Test [Burger is Bad] → Not able to predict new data because it is not present in vocabulary

D1 [[1000000, 0100000, 0010000, 0001000]] 4×7

D2 [[1000000, 0100000, 0010000, 0000100]] 4×7

D3 [[0000010, 0010000, 0000001]] 3×7

Advantage and Disadvantage of OHE

Advantage

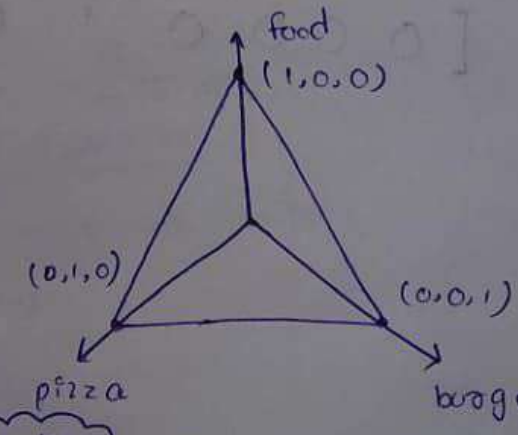
1. Easy to implement with Python
   [Sklearn → One Hot Encoding,
   Pandas → Pd.get_dummies()]

Disadvantage

(150 K vocabulary size)

1. Sparse Matrix → Overfitting
2. ML Algorithm → fixed size i/p
3. No semantic meaning is getting captured.
4. Out of vocabulary (oov)

food	pizza	burger
[1	0	0]
[0	1	0]
[0	0	1]



Note: The distance is almost similar so can't distinguished.